Small Engines Overhaul · Unit 8 · Level 2 (modified)

Engine Teardown & Rebuild



TODAY'S PLAN

- 🧩 Teardown in order
- ⭕ Rings and valves
- 🔧 Reassembly and torque
- 🔍 Troubleshooting and service

Review · 5 minutes

Quick Review



What is the first look-up step in the manual?

Where do you find a part number?

What can void a warranty?

Words to Know

Words to Know



Teardown

Taking an engine apart in order, keeping parts organized and recording each step.



Parts tray

A tray with labeled cups so small parts stay in order during teardown.



Ring end gap

The gap between the ends of a piston ring in the cylinder, checked with a feeler gauge.



Valve clearance

The small gap between the valve stem and the tappet/rocker when the valve is closed.

Words to Know

More Words to Know



🔄 **Lapping valves**: Rubbing a valve on its seat with lapping compound so it seals tightly.

🔧 **Torque**: The twisting force on a bolt, set to spec with a torque wrench (ft-lb or in-lb).

🔍 **Troubleshooting**: Checking fuel, spark, compression and air in order to find why an engine won't run right.

🪢 **Rewind starter**: The pull-rope starter with a coiled spring that rewinds the rope.

Steps

Teardown Order (with adult supervision)



1. Disconnect plug wire; drain fuel and oil
2. Remove shroud, air filter, carburetor, muffler
3. Remove flywheel with a puller; remove armature
4. Remove head, then valves, then piston/rod and crankshaft

Learn

Stay Organized



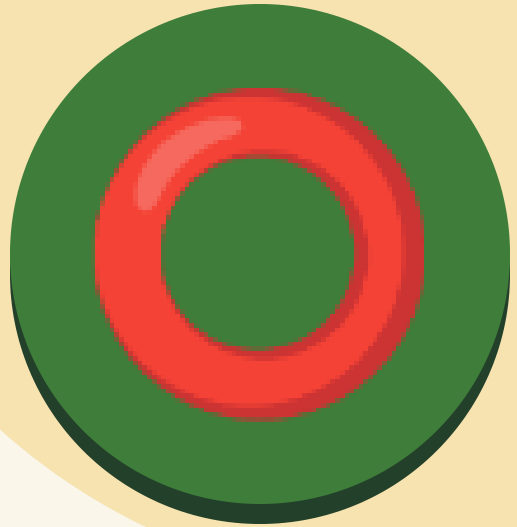
Photo every step; label tray cups in order.

Bag and tag bolts by part.

Keep the manual open to the section you're on.

Learn

Piston Rings

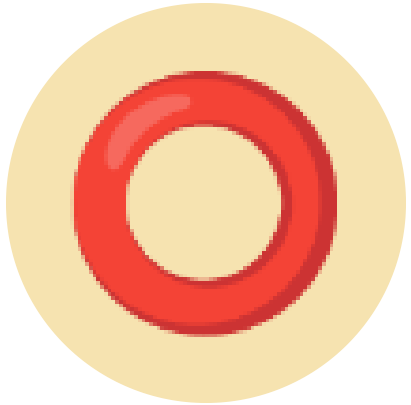


Rings: **top compression**, **second compression**, **oil control**.

End gap: push a ring into the bore, measure the gap with a feeler gauge.

Side clearance: ring in its groove, feeler gauge between ring and land.

🃏 Check 1 of 2



Ring end gap is measured with the ring…

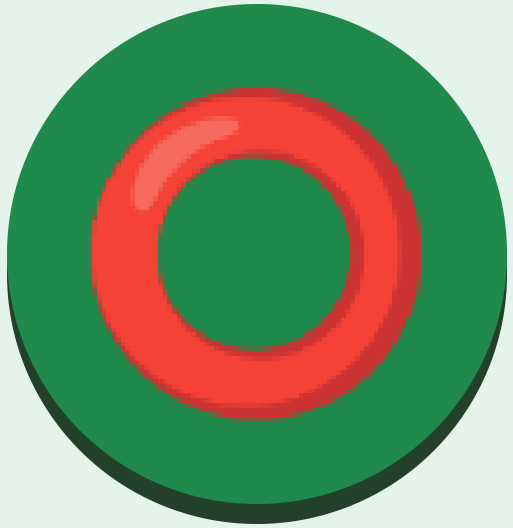
Hold up 1, 2, or 3:

1️⃣ On the floor

2️⃣ Pushed squarely into the cylinder

3️⃣ On the crankshaft

✅ Answer · Card 1 of 2



Pushed squarely into the cylinder

In the bore.

Learn

Valves



Clean carbon; check the face and seat.

Lap with compound until there's an even gray ring.

Set **valve clearance** to spec with a feeler gauge.

Learn

Reassembly



Reverse order; new gaskets; oil moving parts.

Line up **timing marks** on crank and cam gears.

Torque head bolts to spec in the manual's pattern.

Learn

Troubleshooting: Does It Have…



Fuel
fresh, reaching the carb



Spark
blue spark on tester



Compression
gauge to spec



Air
clean filter, choke works

🃏 Check 2 of 2



Spark and compression are good, but no fuel reaches the cylinder. Check the…

Hold up 1, 2, or 3:

1️⃣ Muffler

2️⃣ Flywheel key

3️⃣ Carburetor and fuel line

✅ Answer · Card 2 of 2



Carburetor and fuel line

Fuel path.

Learn

Outdoor Power Equipment Service



Chainsaw: sharp chain, correct tension, bar oil; wear **chaps**, gloves, eye/ear protection.

Trimmer: wind line the right direction; check the drive tube.

Blower / cut-off saw: clean filters; cut-off saws need water or vacuum for silica dust (OSHA).

Wrap-Up

Recap



Teardown
in order, photos



Rings
end gap + side clearance



Valves
lap + clearance



Troubleshoot
fuel, spark, compression, air

Wrap-Up

Exit Ticket



Name the 3 piston rings.

How do you check ring end gap?

List the 4 troubleshooting checks.